

CAREER PROFILE

- Senior process and Operational Excellence engineer with 10 years of experience improving regulated diagnostic, clinical-laboratory, pharmaceutical, medical-device, and high-volume manufacturing workflows through QMS governance, statistical analysis, process validation, automation, and sustainable Lean transformation.
- Led a \$9M Lean Six Sigma program that reduced downtime by 40% and improved final yield by 40%; BD Kaizen portfolio generated approximately \$4M in site impact while reducing deviations by 40% and improving process consistency by 30%.
- Experienced leading change controls, deviations/nonconformances, CAPAs, SOP lifecycle activities, training compliance, audit readiness, process-risk assessments, KPI systems, and corrective-action effectiveness using TrackWise, Veeva, MasterControl, ETQ, SAP, and analytical tools.
- M.S. Mechanical Engineer, DFSS Black Belt, and Lean Six Sigma Green Belt who managed 35 GMP employees and currently coaches 20+ operators and technical partners; skilled in technical mentoring, cross-functional project leadership, data-driven decision-making, and executive communication.

TECHNICAL SKILLS & CREDENTIALS

QMS Governance & Compliance: TrackWise, Veeva, MasterControl, ETQ, change control, document control, SOP lifecycle, deviations, nonconformances, CAPA, NCR, training assignment/completion, on-the-job assessments, quality records, risk registers, metrics, management-review inputs, ISO 13485, ISO 9001 implementation, FDA 21 CFR 800-series and Part 11 awareness, IVDR/IVDD concepts, and CAP/CLIA/NYSDOH/ISO 15189 awareness.

Investigations, CAPA & Regulatory Readiness: Independent investigations, containment, impact assessment, RCA, A3, 8D, 5 Whys, Fishbone, FMEA, fault-tree concepts, evidence packages, audit responses, effectiveness verification, trend analysis, recurring-event prevention, internal/customer/regulatory audit support, inspection readiness, and timely quality-action closure.

Operational Excellence & Lean: Lean Six Sigma, DFSS, DMAIC, Kaizen, 5S/6S, value-stream and process mapping, current/future-state design, standard work, mistake proofing, visual management, tiered Daily Management, flow, constraint analysis, workload leveling, takt/cycle time, waste elimination, change adoption, pilot/measure/scale execution, and benefit sustainment.

Operational Data, KPI & Statistical Analytics: Power BI, Excel, JMP, Minitab, OSI-PI, MES, SAP, Smartsheet; dashboard and scorecard development, automated reporting concepts, SPC/QC, control charts, DOE, capability, MSA/Gage R&R, regression, Pareto, trend and variation analysis, leading indicators, data integrity, KPI definition, backlog/ticket tracking, prioritization, and management reporting.

Diagnostics & Laboratory Operations: High-complexity diagnostic and microbiological operations support, molecular/NGS workflow awareness, pre-analytical, analytical and post-analytical process concepts, specimen/material flow, media preparation, laboratory equipment, sample/data traceability, automated inspection, testing workflows, turnaround-time and capacity improvement, quality events, and regulated clinical-operations collaboration.

Process Engineering, Validation & Automation: Process design and monitoring, validation strategy, IQ/OQ/PQ, test-method validation, verification protocols, equipment qualification, PLC/HMI, FactoryTalk View, machine vision, automated equipment, process controls, maintenance/calibration systems, SAP/CMMS, technical troubleshooting, process requirements, technology transfer, risk-based change implementation, and sustained operational support.

Documentation, Training & Standardization: SOPs, work instructions, guidelines, verifications, protocols, reports, process flows, technical proposals, cost-saving recommendations, controlled templates, document obsolescence prevention, training matrices, competency assessments, adult-learning methods, technical mentoring, process standardization, and point-of-use documentation.

Project & Stakeholder Leadership: Concurrent workstream management, charters, scope, schedules, budgets up to \$4M, resource plans, risks/issues, structured backlogs, governance reviews, cross-functional prioritization, technical requirements translation, Engineering/Software/Product/Operations collaboration, executive presentations, influencing without authority, and timely project delivery.

PROFESSIONAL EXPERIENCE

Process Engineer - Operations | BD Integrated Diagnostics | Mebane, NC | Nov 2024 – Present

- Generated approximately \$4M in site impact by leading 2 Kaizen events and co-leading 1 that reduced media-preparation deviations by 40% and improved process consistency by 30%.
- Lead QMS, process engineering, Operational Excellence, validation, automation, laboratory/diagnostic operations, reliability, supplier, and continuous-improvement initiatives for high-volume FDA-regulated products and workflows.
- Own and support change controls, deviations/nonconformances, CAPAs, investigations, document revisions, SOPs, work instructions, training workflows, risk assessments, approvals, effectiveness checks, and inspection-ready records using TrackWise, Veeva, MasterControl, and ETQ.
- Independently investigate process, equipment, software, material, documentation, and human-factor events using containment, impact assessment, data review, RCA, FMEA, A3, 8D, 5 Whys, Fishbone, DOE, and controlled testing.
- Develop KPI dashboards and scorecards using Power BI, SAP, MES, OSI-PI, Excel, JMP, Minitab, and Smartsheet to monitor throughput, cycle time, yield, deviations, CAPA/NC aging, quality, training, capacity, equipment performance, and improvement benefits.

- Apply SPC, capability analysis, MSA, trend analysis, and leading-indicator monitoring to identify variation, emerging process risks, undesirable events, and proactive interventions before performance or compliance is affected.
- Partner with Quality Assurance, Regulatory, Clinical/Scientific Operations, Validation, Engineering, Automation, Maintenance, Supply Chain, Software/data partners, vendors, and leadership to translate operational needs into technical requirements and controlled solutions.
- Develop and maintain SOPs, work instructions, process maps, technical reports, validation records, guidelines, training materials, OJT assessments, evidence packages, and audit-observation responses aligned with ISO 13485 and FDA-regulated QMS expectations.
- Manage concurrent process, quality, equipment, and automation projects with scopes, schedules, direct budgets up to \$4M, risks, suppliers, milestones, validation deliverables, governance reviews, operational handoff, and management recommendations.
- Train and mentor 20+ operators and technical partners in workflows, QMS processes, regulatory expectations, process controls, troubleshooting, data interpretation, Lean methods, and standardized execution.

Process Engineer - Continuous Improvement Operations | Pfizer | Kalamazoo, MI | Oct 2021 - Oct 2024

- Reduced downtime by 40% and improved final yield by 40% by leading a \$9M Lean Six Sigma program using DOE, SPC, capability analysis, equipment/process improvements, Power BI, JMP, Minitab, MES, and historian data.
- Managed 35 GMP employees in 24/7 operations; established priorities, assigned work, maintained training, developed colleagues, conducted performance discussions, resolved escalations, and balanced safety, quality, delivery, cost, compliance, and engagement.
- Implemented Pfizer IMEx, tiered Daily Management, visual controls, standard work, process confirmation, flow improvements, structured problem solving, deviation reduction, and employee capability development across complex regulated workflows.
- Built and maintained dashboards for production, quality, deviations, reliability, capacity, cycle time, schedule, training, and project benefits; interpreted trends and led cross-functional action reviews.
- Led deviations, CAPAs, change controls, investigations, SOP and batch-document revisions, validation impact assessments, training, audit readiness, process/equipment risk mitigation, and recurring-event prevention.
- Partnered across Manufacturing, Quality, Engineering, Validation, laboratories, Automation, Maintenance, Supply Chain, data-system teams, vendors, and leadership to pilot, measure, scale, and sustain process improvements.
- Authored technical reports, procedures, risk assessments, training materials, business cases, project updates, and executive recommendations translating complex operational data into actionable decisions.

Engineer - Operations | Maini Materials Movement Ltd. | India | May 2016 - Aug 2019

- Improved manufacturing efficiency from 68% to 92%, reduced molding cycle time by 18%, and lowered scrap by 30% through process mapping, line balancing, standard work, supplier development, visual management, and 6S.
- Analyzed process flow, cycle time, capacity, equipment constraints, material movement, inspection data, defects, and customer requirements to develop future-state processes, controls, and measurable improvement plans.
- Led cross-functional and supplier initiatives involving SOPs, PFMEA, Control Plans, SPC, MSA, validation, 8D corrective action, engineering changes, training, production readiness, and sustained launch performance.
- Supported ERP/MRP, production scheduling, structured issue tracking, material readiness, equipment availability, quality metrics, and customer-delivery recovery plans.
- Led suppliers and cross-functional teams through technical specifications, quotations, design/manufacturability reviews, APQP/PPAP, FAI, capability, equipment trials, corrective actions, maintenance readiness and stable production.
- Developed and supported mechanical/electromechanical systems using sensors, actuators, PLC/HMI concepts, automated equipment, fixtures, tooling, utilities, drawings, GD&T, DFM/DFA, PFMEA, Control Plans, commissioning, validation and production ramp-up.

Team Leader - Vehicle Dynamics, Powertrain, and Chassis | Chicane Racing | India | May 2014 - May 2016

- Led a multidisciplinary team through project planning, requirements, resource allocation, technical reviews, fabrication, assembly, testing, schedules, budgets, risks, presentations, coaching, and final delivery.
- Validated 15% fuel economy improvement through accelerated warm-up analysis of powertrain fluids across engine, transmission, and differential systems.
- Supported prototype design reviews, hardware testing, validation planning, troubleshooting, technical decision-making, ECU validation, rapid control prototyping, hardware-in-the-loop simulation, measurement/diagnostics, and automatic transmission testing.

EDUCATION & CERTIFICATIONS

M.S., Mechanical Engineering | University of South Florida, Tampa, FL

B.Tech., Mechanical Engineering | SRM Institute of Science and Technology, India

DFSS Black Belt Certification | Lean Six Sigma Green Belt | TPM | DMAIC